

Given a sequence of n flips with k "Heads"

The probability it was generated by a fair coin is

$$\text{For rigid it's } \left(\frac{1}{2}\right)^n$$
$$\left(\frac{3}{4}\right)^k \cdot \left(\frac{1}{4}\right)^{n-k}$$

We can calculate a confidence score

Casino Problem

Input: Given a sequence of coin flips, determine whether the dealer used a fair coin and a biased coin, and the probabilities as well.

Output: Which coin used at each position

We can consider windows and decide the best coin in the window

HMM

The symbols emitted are either H and T

There is a set of hidden states F and B

There are also transition probabilities

And emission probabilities

An observer can see the emitted symbols of an HMM but does not know the specific state the HMM is in,

Since x is fixed, we wish to compute $\Pr(x, \pi)$ maximized over all π

$$\Pr(x, \pi) = \Pr(\pi) \cdot \Pr(x | \pi)$$

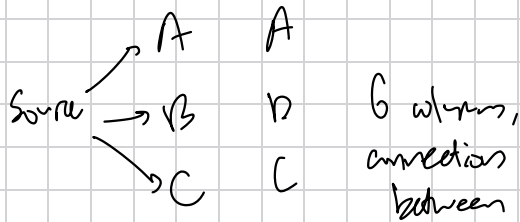
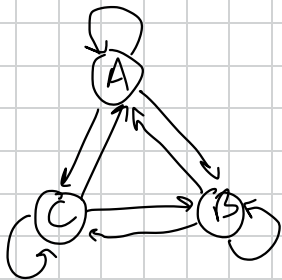
Decoding Problem

Input: A string $x = x_1, \dots, x_n$ emitted by an HMM
(Σ , States, Transition, Emission)

Output: A path π that maximizes the probability $P(x, \pi)$ over all paths

This is solvable with DP on a DAG

The DAG is known as the Viterbi Graph



The transitions are just the product of moving to that state and emitting the correspondingly symbol

Then we just DP on DAG

Original Problem (Multiple Alignments)

Scab Alignment

If 40% of symbols is -, then remove the column

Then we make the profile matrix of the alignment